



Research article

FedMamba: Robust multimodal federated intrusion detection for heterogeneous IoT systems

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ABSTRACT

The convergence of Information Technology (IT) and Operational Technology (OT) in Industry 4.0 produces diverse data streams, such as system logs, sensor readings, and network traffic, which are vital for industrial security. However, existing security analytics are siloed by modality and rely on centralized processing, raising concerns regarding privacy, latency, and scalability. Although Federated Learning (FL) mitigates privacy risks, most frameworks remain unimodal, lack support for non-IID data distributions, and face adversarial evasion challenges. We propose FedMamba, a novel multimodal Federated Learning (MMFL) framework that creates a unified Mamba-based model to address these issues via (i) efficient cross-modal learning, (ii) a FedProx-based protocol for stable non-IID training that remains compatible with secure aggregation, and (iii) modality-specific adversarial training for robustness. Experiments on HDFS, SWaT, and CICIOT-2024 datasets show that the standard FedMamba achieved competitive macro F1-scores of 0.9584, 0.9795, and 0.9665 relative to centralized baselines, but degraded on HDFS and SWaT under PGD attack (0.3791 and 0.5147), whereas CICIOT-2024 remained robust under the same attack (0.9665). The adversarially trained FedMamba-AT sustains robust F1-scores (0.9480, 0.8357, 0.9645). FedMamba offers a robust and scalable solution for secure IIoT monitoring.

1. Introduction

The fourth industrial revolution, or Industry 4.0, represents the convergence of cyber-physical systems and advanced analytics in manufacturing environments, driven by the Industrial Internet of Things (IIoT) [1]. This paradigm empowers smart factories to generate voluminous and heterogeneous data streams that span system logs, sensor telemetry, and network activity. These data are vital for operational efficiency and real-time threat detection. However, the interconnected nature of Operational Technology and Information Technology systems creates complex interdependencies, where anomalies in one domain may signal or cause disruptions in another. This calls for a unified multimodal security analytics framework that is capable of cross-domain inference.

Conventional security analytics are insufficient to address these challenges. Typically, independent models are deployed for each data modality, resulting in siloed insights and a high operational overhead. Centralized analytics systems attempt to fuse data centrally but face bandwidth bottlenecks, privacy regulations, and single points of failure. Consequently, there is a growing need for holistic and decentralized solutions that preserve privacy while leveraging the full spectrum of heterogeneous factory data.

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FL has emerged as a promising paradigm that trains a global model collaboratively without aggregating raw data [2,3]. However, deploying FL in real-world industrial environments poses several challenges. First, clients naturally possess statistically heterogeneous (non-IID) data, which can cause instability and failure to converge in standard FL algorithms [4]. Second, the transmission of model updates to a central server, even without raw data, creates privacy vulnerabilities against an “honest-but-curious” server [5]. Third, early studies commonly assumed that all clients possessed homogeneous, single-modality data, which is an unrealistic assumption in diverse IIoT environments [6].

This limitation has motivated the development of multimodal federated learning (MMFL), in which clients can hold different data modalities. Surveys such as [6] and recent advances in MMFL [7–9] have explored architectural solutions for modality heterogeneity; however, these methods are primarily evaluated on benign, non-security datasets (e.g., image-text benchmarks) and do not evaluate the adversarial threat landscape of industrial environments.

Federated models are susceptible to inference-time adversarial attacks, which can subvert anomaly detection and classification [10]. Although defenses such as adversarial training have been studied in homogeneous FL scenarios [11], their extension to multimodal environments is non-trivial. Each modality requires custom perturbation strategies, and unified robustness is not achieved by naïve adaptation.

The intersection of multimodality, decentralization, and adversarial robustness remains largely unaddressed, despite its significance in IIoT systems. Surveys and representative MMFL methods [6–9] primarily evaluate benign, non-security benchmarks and do not address adversarial robustness or extreme modality and feature skew typical of smart factories. In parallel, FL security works on inference-time attacks and adversarial training [10,11] and generally assumes a single modality, leaving modality-specific perturbation models and aggregation under modality skew unexamined. Prior FL work for IIoT/ICS [12–16] also omits an end-to-end adversarially robust multimodal design.

To address this gap, we propose FedMamba, a framework that jointly addresses data heterogeneity, training stability, privacy, and adversarial robustness in IIoT security. The framework unifies system logs (HDFS), sensor time series (SWaT), and network flows (CICIoMT-2024) using a shared Mamba backbone. Its core is a robust federated protocol that leverages FedProx to promote stable convergence on non-IID data and incorporates a secure-aggregation simulation to protect clients’ updates during aggregation. Finally, it integrates a modality-specific adversarial training scheme to harden the final model against inference-time attacks [6–11]. Our framework, to our knowledge, is among the first to jointly address multimodal heterogeneity, privacy, and adversarial robustness in IIoT.

Our key contributions are:

- 1) **Unified architecture for modality feature skew.** We design a single multimodal model with explicit parameter partitioning $\theta = \theta_{\text{in}} \cup \theta_{\text{back}} \cup \theta_{\text{cls}}$. Modality-specific adapters and heads isolate heterogeneity across logs, time series, and tabular flows, while a shared Mamba backbone enables cross-modal transfer with minimal interference (Fig. 2 and Section 3.3).
- 2) **Modality-aware block-wise aggregation compatible with secure aggregation.** Under one-modality-per-client, we provide a principled weighting scheme that averages the shared backbone across all clients and averages each modality block only within that modality. This is implemented by client-side pre-scaling and server-side partitioned averaging, which fits standard secure aggregation without protocol changes (Eqs. (15)–(17) and Sections 3.4, 3.5).
- 3) **Federated optimization under modal and statistical heterogeneity.** We justify why FedProx’s non-convex analysis extends to our block-partitioned model because each block receives an unbiased estimate of its sub-objective under the proposed aggregation. Empirically, this prevents the collapse observed with FedAvg under severe non-IID skew (Algorithm 1 and Section 4).
- 4) **Selective federated adversarial training with cross-modal robustness transfer.** We introduce a selective scheme that adversarially trains only vulnerable modalities, and show that the robustness learned in the shared backbone transfers across modalities. This preserves clean accuracy on non-trained modalities and reduces client-side cost compared with training all modalities (Section 3.6 and Table 3).

FedMamba addresses heterogeneous IIoT environments with a shared Mamba backbone, modality-aware block-wise aggregation that remains compatible with secure aggregation, FedProx-based optimization, and selective adversarial training.

The remainder of the paper is organized as follows. Section 3 presents the system and threat model, the FedMamba architecture, training protocol, secure aggregation compatibility, and selective adversarial training. Section 4 covers experiments, robustness analyses, ablations, and scalability. Section 5 concludes, and Appendices A–D provide supporting details.

2. Related work

Our research is situated at the confluence of three critical domains: FL for IIoT security, the emerging field of MMFL, and adversarial robustness in decentralized systems. We reviewed the state-of-the-art in each domain to precisely situate the contributions of our FedMamba framework.

FL is widely used as a privacy-preserving approach for security analytics in IIoT. A significant body of work has focused on applying FL to network intrusion detection in unimodal settings. Foundational frameworks such as DeepFed [12] demonstrate the efficacy of using federated CNN-GRU models to achieve high detection rates on network traffic data while preserving privacy. This line of research has been extended through various architectural and algorithmic innovations in recent years. For instance, some works have explored hybrid ensembles [13], federated transfer learning [15], and optimized hybrid models like CNN-LSTM [16] to improve generalization and accuracy.

Given the edge resource constraints, several studies have emphasized efficiency, including lightweight designs for edge deployment [14]. Concurrently, a parallel stream of research has addressed the foundational challenges of the federated process. A primary issue is statistical heterogeneity, where non-IID client data can cause model divergence ("client drift"), leading to unstable convergence. Foundational algorithms, such as FedProx [4], were developed to counteract this by introducing a proximal regularization term. Another critical challenge is the privacy of model updates, as a curious server can potentially infer client data. Cryptographic protocols, such as secure aggregation, allow the server to aggregate only masked updates [5]. These solutions are typically evaluated in a unimodal context.

Despite these significant advancements in architecture, efficiency, and FL protocol security, a common thread unites this body of work: it operates under the assumption that all clients possess homogeneous, single-modality data. These frameworks, while powerful, are predominantly designed for and evaluated on a single data type, such as network traffic [12,13,15–18] or sensor data [14]. However, they do not address the intrinsic heterogeneity of smart factories, where system logs, sensor telemetry, and network traffic coexist.

The challenge of learning from diverse data modalities across clients has given rise to multimodal federated learning (MMFL). A survey by Che et al. [6] provides a foundational taxonomy of architectural challenges, including client-level modality heterogeneity. Recent studies have proposed specific architectural solutions, such as enabling clients with partial modalities to collaborate [7], using contrastive learning to align representations [9], or employing multimodal transformers [8]. Although architecturally innovative, these methods are primarily validated on benign, non-industrial benchmarks (such as vision-language or medical datasets), with little or no evaluation of IIoT security data. Furthermore, they generally do not incorporate adversarial threat models critical to security applications, nor do they typically address the challenge of ensuring stable convergence under the dual pressures of modal and statistical (non-IID) heterogeneity.

Federated systems are vulnerable to adversarial attacks. Research in this area has explored the factors influencing robustness [10] and proposed methods for propagating robustness between clients [11], typically in unimodal and homogeneous scenarios. From an industrial perspective, Kim [19] provides an insightful analysis of adversarial attacks in centralized industrial control systems (ICS) environments, while Ahmad et al. [20] propose a StuxNet-inspired adversarial learning framework for detecting advanced threats in critical infrastructure. However, a significant gap remains in the development of modality-aware adversarial defense strategies within a federated context, which is a challenge central to our work.

To unify advancements across these domains and address their limitations, we propose FedMamba, an MMFL framework explicitly designed for industrial IoT security. Unlike traditional FL for IoT, our approach is designed to handle client-level modality heterogeneity. In contrast to prior MMFL work, we integrate adversarial robustness as a first-class design objective and ensure the privacy of updates via secure aggregation. Finally, our approach combines FedProx for stability under non-IID multimodal data with modality-aware adversarial training, providing a practical end-to-end solution for heterogeneous smart factory environments.

3. Methodology

In this section, we formalize FedMamba and its training protocol. We first establish centralized unimodal baselines to calibrate the expected performance ceiling. We then specify the unified multimodal architecture, together with a FedProx-based objective and a modality-aware aggregation rule that is compatible with secure aggregation. Finally, we extend the protocol with modality-specific adversarial training to improve inference-time robustness. Fig. 1 provides an overview; Sections 3.2, 3.3, and 3.6 present the corresponding definitions, equations, and algorithms. The complete notation reference is provided in Appendix A (Table A.1).

3.1. System and threat model

We define the system architecture and threat model to formally scope the security and privacy contributions.

3.1.1. System model

We model a distributed system, such as a smart factory, which comprises two primary entities:

3.1.1.1. Clients. A set of K heterogeneous clients, $\mathcal{K} = \{1, \dots, K\}$, where each client $k \in \mathcal{K}$ possesses a private local dataset $\mathcal{D}_k$ corresponding to exactly one modality $m_k \in \mathcal{M}$. Clients have sufficient computational resources for local training but are constrained from sharing raw data because of privacy or network-related limitations.

3.1.1.2. Central server. A single server orchestrates this federated process. We assume an honest-but-curious coordinator. To protect client updates, we simulate a secure-aggregation setting [5] to evaluate compatibility (see Section 3.5).

3.1.2. Threat model

We assess the robustness of the trained global model $f(\cdot; \theta)$ to inference-time evasion attacks.

3.1.2.1. Adversary's goal. The adversary's goal is to evade detection. For a given input x with true label y , the adversary seeks to craft a new input $x' = x + \delta$ by adding a small perturbation δ , such that the model's prediction is incorrect, that is, $\arg \max_{c \in \{1, \dots, C\}} [f(x'; \theta)]_c \neq y$.

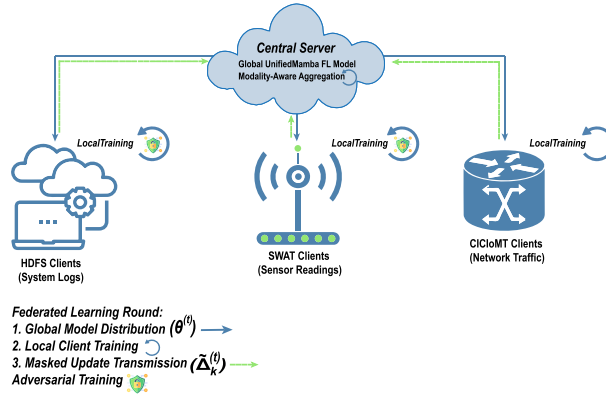


Fig. 1. System architecture of FedMamba. A central server orchestrates federated learning across heterogeneous clients processing different modalities: HDFS (system logs), SWaT (sensor time series), and CICIoMT-2024 (network traffic). In each round t : (1) the server broadcasts the global model $\theta^{(t)}$ to selected clients; (2) each client trains locally on private data; clients marked with a shield (HDFS, SWaT) also run adversarial training, while CICIoMT-2024 uses standard training; (3) clients transmit masked, sample-weighted updates; (4) the server applies modality-aware aggregation, averaging backbone parameters over all clients (weighted by N_t) and modality-specific parameters only within the same modality (weighted by $N_t^{(m)}$).

3.1.2.2. Adversary's knowledge. We evaluate two adversary knowledge assumptions: (i) *white-box*, where the adversary knows the architecture, parameters θ , and preprocessing, and can compute exact input gradients (worst-case evaluation [21]), and (ii) *transfer-based black-box*, where the adversary trains a surrogate model and crafts ℓ_∞ -bounded perturbations on the surrogate that are applied to the target without queries.

3.1.2.3. Adversary's capabilities. The primary threat is inference-time evasion: the adversary perturbs inputs with ℓ_∞ budget ϵ (i.e., $\|\delta\|_\infty \leq \epsilon$), clipped to valid feature ranges when applicable. The adversary cannot influence the training data, objectives, clients, or aggregation. Training-time threats (poisoning, backdoors, and Byzantine faults) are out of scope for the core method; nevertheless, we report a separate backdoor experiment to quantify baseline susceptibility without altering the training protocol (Appendix D).

3.1.3. Training-time backdoor probe

To quantify the baseline susceptibility, we ran a short training-time backdoor poisoning probe without changing the learning rule or secure aggregation simulation. In each round, 12% of the selected clients were malicious. A malicious client poisons a batch with a probability of 0.25 and modifies a random 30% of the examples in that batch, relabeling them to the target class. Triggers are modality-aware: HDFS sets the last two tokens to id ($|\mathcal{V}| - 1$); SWaT adds +0.015 to the first two sensors over the last eight time steps; CICIoMT-2024 adds +0.010 to the first three features. Other knobs match our main setup: FedProx $\mu=0.01$, one local epoch, client fraction 0.4, Dirichlet $\alpha=0.3$. We report clean Accuracy and Macro F1 on unmodified test data and Attack Success Rate (ASR) over non-target examples with the trigger. The results are summarized in Table D.1.

3.2. Foundational unimodal models

To establish rigorous performance baselines, we developed three independent centralized models, each of which was specialized for a single modality. These unimodal models served as non-federated performance benchmarks. Each model follows the partitioned structure of the final federated architecture, comprising a modality-specific input adapter ($\theta_{in,m}$), a Mamba backbone (θ_{back}), and a task-specific classifier head ($\theta_{cls,m}$).

3.2.1. mamba for HDFS log classification (M_{HDFS})

The HDFS dataset [22] contains tokenized log sequences of length $L_H = 50$. The model uses an embedding layer as its input adapter ($f_{in,H}$), processes the sequence with the Mamba backbone, and classifies the final hidden state. The anomaly probability is:

$$p(y=1 | x) = \sigma \left(f_{cls,H} \left(\left[\text{Mamba} \left(f_{in,H}(x; \theta_{in,H}); \theta_{back} \right) \right]_{L_H} \right) \right) \quad (1)$$

Training employed the AdamW optimizer with a class-weighted binary cross-entropy (BCE) loss.

Table 1
Overview of datasets used for evaluation.

Dataset	Modality	# Samples	# Features	Task
HDFS	System Logs	575,061	29 (Events)	Binary Anomaly
SWaT	ICS Sensors	449,919	51	Binary Attack
CICIoMT-2024	Network Traffic	6,957,003	39	Multiclass Attack

3.2.2. Mamba for SWaT sensor attack classification (M_{SWaT})

The SWaT dataset [23] provides time-series data segmented into windows of $L_S = 60$. A linear projection serves as the input adapter ($f_{in,S}$). The model computes,

$$p(y=1 | x) = \sigma \left(f_{cls,S} \left(\left[\text{Mamba}(f_{in,S}(x; \theta_{in,S}); \theta_{back}) \right]_{L_S} \right) \right) \quad (2)$$

Training employed AdamW with class-weighted BCE.

3.2.3. Mamba for CICIoMT-2024 network traffic classification (M_{CIC})

The CICIoMT-2024 dataset [24] contains network-flow features treated as single-step sequences. The architecture uses a projection layer ($f_{in,C}$), layer normalization, and two stacked Mamba blocks (collectively parameterized by θ_{back}), followed by a classifier head. Let,

$$s = f_{cls,C}(\text{Mamba}_2(\text{Mamba}_1(\text{LN}(f_{in,C}(x; \theta_{in,C}))))), \quad (3)$$

where $s \in \mathbb{R}^C$ is the logit for C attack classes. The predicted probability of class c is then

$$p(y=c | x) = \frac{\exp(s_c)}{\sum_{j=1}^C \exp(s_j)}. \quad (4)$$

Training used AdamW with class-weighted cross-entropy (CE) loss and gradient clipping. An overview of the datasets used for evaluation is provided in Table 1.

3.3. A robust multimodal federated learning framework with simulated secure aggregation

The central challenge is learning from clients with fundamentally different data modalities, an extreme form of non-IID data distribution that we term *modality feature skew*, where the feature space and data structure differ across client types. Two further requirements for deployment are robustness to statistical heterogeneity (unbalanced labels within a modality) and data privacy preservation. We address this threefold challenge by co-designing a unified model architecture and a robust, privacy-aware, federated training algorithm.

3.3.1. The unified multimodal architecture

The architecture of the unified model is illustrated in Fig. 2, and the model is designed with an explicit parameter partition to address modality feature skew. The global parameters θ are partitioned into three disjoint sets: a shared feature-extraction backbone θ_{back} , modality-specific input adapters θ_{in} , and classifier heads θ_{cls} . Let $\mathcal{M} = \{\text{HDFS}, \text{SWaT}, \text{CICIoMT-2024}\}$ be the set of modalities. Modality-specific parameter sets are formed by the union of parameters for each modality, where the complete model is the disjoint union of these partitions, denoted by ϖ :

$$\begin{aligned} \theta_{in} &= \bigcup_{m \in \mathcal{M}} \theta_{in,m}, & \theta_{cls} &= \bigcup_{m \in \mathcal{M}} \theta_{cls,m}, \\ \theta &= \theta_{in} \uplus \theta_{back} \uplus \theta_{cls}. \end{aligned} \quad (5)$$

For readability, we present only the essential symbols; a complete notation table is provided in Appendix A.

3.3.1.1. Modality-specific input layers (θ_{in}). We project heterogeneous raw inputs into a common latent space using non-shared adapters with parameters $\theta_{in,m}$ for each modality $m \in \mathcal{M}$.

$$\mathbf{H}_0^{(H)} = f_{in,H}(x^{(H)}; \theta_{in,H}) \in \mathbb{R}^{L_H \times d_{\text{model}}}, \quad (6)$$

$$\mathbf{H}_0^{(S)} = f_{in,S}(x^{(S)}; \theta_{in,S}) \in \mathbb{R}^{L_S \times d_{\text{model}}}, \quad (7)$$

$$\mathbf{H}_0^{(C)} = f_{in,C}(x^{(C)}; \theta_{in,C}) \in \mathbb{R}^{L_C \times d_{\text{model}}}. \quad (8)$$

In our implementation $L_C = 1$ for tabular CICIoMT-2024 samples, $L_S = 60$ for SWaT sliding windows, and $L_H = 50$ for HDFS event sequences. This partitioned input design addresses the modality feature skew by dedicating a separate, non-shared adapter per modality, enabling each data type to learn an optimal transformation without interference and support meaningful cross-modal learning.

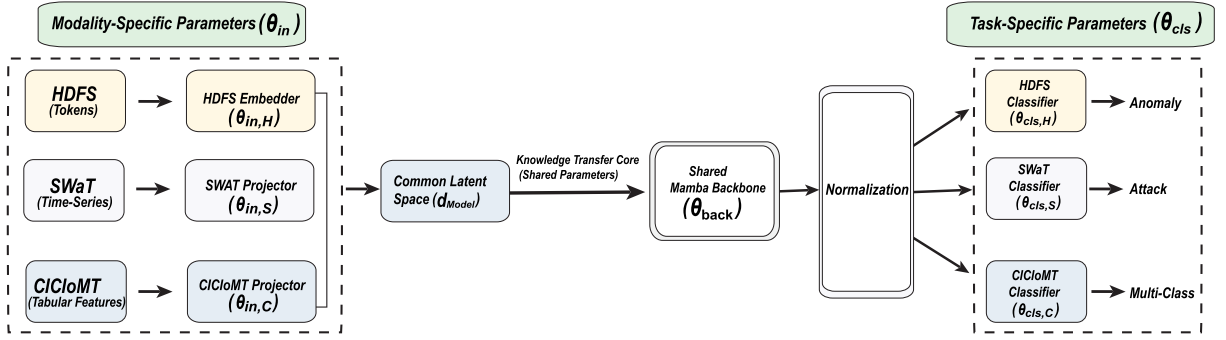


Fig. 2. Unified multimodal model with explicit parameter partition. Input: heterogeneous data were projected by non-shared adapters ($\theta_{in,H}$, $\theta_{in,S}$, $\theta_{in,C}$) into a shared latent space of dimension $d_{model}=256$ (HDFS $L_H=50$, SWaT $L_S=60$, CICIoMT-2024 $L_C=1$). Shared processing: A single Mamba backbone (θ_{back}) enables cross-modal feature transfer with linear complexity $\mathcal{O}(L_m)$. Output: task-specific heads ($\theta_{cls,H}$, $\theta_{cls,S}$, $\theta_{cls,C}$) produced predictions for anomaly detection (HDFS), binary attack detection (SWaT), and multiclass traffic classification (CICIoMT-2024).

3.3.1.2. Shared Mamba backbone (θ_{back}). A shared Mamba selective state-space model [25], parameterized by θ_{back} , processes the embedded sequences as follows:

$$\mathbf{H}_{back}^{(m)} = \text{Mamba}(\mathbf{H}_0^{(m)}; \theta_{back}) \in \mathbb{R}^{L_m \times d_{model}}. \quad (9)$$

We take the representation at the last time step and apply the layer normalization. Writing $\mathbf{H}_{back}^{(m)} = [\mathbf{h}_1^{(m)}; \dots; \mathbf{h}_{L_m}^{(m)}]$ with $\mathbf{h}_t^{(m)} \in \mathbb{R}^{d_{model}}$, we set

$$\mathbf{z}^{(m)} = \text{LN}(\mathbf{h}_{L_m}^{(m)}) \in \mathbb{R}^{d_{model}}, \quad (10)$$

where $\text{LN}(\cdot)$ denotes LayerNorm.

The Mamba backbone processes sequences with computational complexity $\mathcal{O}(B L_m d_{model} d_{state})$ per forward pass for batch size B , which scales linearly with sequence length L_m . This contrasts with the quadratic complexity of transformer self-attention, $\mathcal{O}(B L_m^2 d_{model})$.

3.3.1.3. Task-specific classifier heads (θ_{cls}). Independent classifier heads, with parameters $\theta_{cls,m}$ for each modality $m \in \mathcal{M}$, map the shared representation to task-specific outputs as follows:

$$\hat{y}^{(m)} = f_{cls,m}(\mathbf{z}^{(m)}; \theta_{cls,m}). \quad (11)$$

Maintaining separate classifiers preserves modality-specific decision criteria while enabling shared representation learning.

3.3.1.4. Task losses and class imbalance. For the binary classification tasks (HDFS and SWaT), we employed class-weighted binary cross-entropy (BCE) loss. Given a logit s and a label $y \in \{0, 1\}$, the loss is:

$$\ell_{\text{BCE}}(s, y) = -(w_+ \cdot y \log \sigma(s) + w_- \cdot (1 - y) \log(1 - \sigma(s))), \quad (12)$$

where $\sigma(\cdot)$ is the sigmoid function and w_+ , w_- are weights for the positive and negative classes, respectively, to counteract class imbalance. For the Multiclass CICIoMT-2024 task, we used a standard cross-entropy (CE) loss. Given a vector of logits $\mathbf{s} \in \mathbb{R}^C$ for C classes and a ground-truth label y , the loss is:

$$\ell_{\text{CE}}(\mathbf{s}, y) = -\log \left(\frac{\exp(s_y)}{\sum_{c=1}^C \exp(s_c)} \right). \quad (13)$$

Class weights were applied by scaling each sample according to the class frequency.

3.4. Robust federated training protocol

Training our multimodal architecture in a non-IID and privacy-sensitive environment requires addressing two primary challenges: (i) instability caused by statistical heterogeneity and (ii) the need for modality-aware aggregation that respects the partitioned parameter structure. We address these issues using a principled federated training protocol built on FedProx with custom aggregation rules.

3.4.1. FedProx for non-IID stability

Standard Federated Averaging (FedAvg) implicitly assumes that each client's local objective is a good proxy for the global objective. However, under non-IID data distributions, where clients hold highly skewed subsets of classes (Fig. 3), this assumption fails. Unconstrained local training causes client model weights to drift toward solutions that are optimal for locally biased data but suboptimal globally, leading to unstable convergence.

We employed FedProx [4], which introduces a proximal regularization term to constrain local models near the global model. Each client k minimizes the following:

$$\mathcal{J}_k(\theta_k) = \mathcal{L}_k(\theta_k; \mathcal{D}_k) + \frac{\mu}{2} \|\theta_k - \theta^{(t)}\|_2^2, \quad (14)$$

where $\mathcal{L}_k$ is the local loss on data $\mathcal{D}_k$, $\theta^{(t)}$ is the global model from round t , and μ is the regularization strength. In our architecture, local optimization updates only parameters active for client k 's modality m_k : the shared backbone (θ_{back}) and modality-specific layers ($\theta_{\text{in},m_k}, \theta_{\text{cls},m_k}$).

Under non-IID conditions, the proximal term acts as an elastic anchor: clients adapt to local data while large deviations from the global model $\theta^{(t)}$ are being penalized. The parameter μ controls the anchor strength: a larger μ enforces tighter adherence to the global model, whereas a smaller μ allows greater local flexibility.

3.4.2. Modality-aware block-wise aggregation

A key challenge in our multimodal framework is the principled aggregation of updates from clients with different modalities. Standard weighted averaging is insufficient because different parameter blocks serve different purposes: the backbone learns cross-modal features, whereas modality-specific blocks learn specialized transformations.

We average the shared backbone across all participating clients (weighted by the total samples N_t), enabling it to learn modality-agnostic representations. For modality-specific blocks (input adapters, classifier heads), we average only over clients of that modality (weighted by $N_t^{(m)}$), preserving the specialization. Formally:

$$\Delta_{\text{back}}^{(t+1)} = \frac{1}{N_t} \sum_{k \in S_t} n_k \Delta_{\text{back},k}^{(t)}, \quad (15)$$

$$\Delta_{\text{in},m}^{(t+1)} = \frac{1}{N_t^{(m)}} \sum_{k \in S_t^{(m)}} n_k \Delta_{\text{in},m,k}^{(t)}, \quad \forall m \in \mathcal{M}, \quad (16)$$

$$\Delta_{\text{cls},m}^{(t+1)} = \frac{1}{N_t^{(m)}} \sum_{k \in S_t^{(m)}} n_k \Delta_{\text{cls},m,k}^{(t)}, \quad \forall m \in \mathcal{M}, \quad (17)$$

where $n_k = |\mathcal{D}_k|$, $N_t = \sum_{k \in S_t} n_k$, and $N_t^{(m)} = \sum_{k \in S_t^{(m)}} n_k$. The global model updates component-wise: $\theta_{\text{part}}^{(t+1)} \leftarrow \theta_{\text{part}}^{(t)} + \Delta_{\text{part}}^{(t+1)}$.

Implementation via pre-scaling. This differential weighting is implemented through client-side pre-scaling: each client multiplies its update $\Delta_k^{(t)}$ by its sample count n_k before the transmission. The server receives scaled updates $\{n_k \Delta_k^{(t)}\}$, sums them, then divides the backbone components by N_t and modality blocks by $N_t^{(m)}$. This design is fully compatible with secure aggregation, as detailed in Section 3.5.

3.4.3. Convergence analysis

We rely on the established FedProx convergence guarantees [4] for non-convex federated optimization. Let the global objective be $f(\theta) = \sum_{k=1}^K p_k F_k(\theta)$ with $p_k = n_k / \sum_j n_j$, where F_k is client k 's local objective. Under standard assumptions: (i) each F_k is L -smooth; (ii) local dissimilarity is bounded; and (iii) local solutions are γ -inexact with a proximal parameter $\mu > 0$. Under these conditions, FedProx guarantees a sufficient per-round decrease:

$$\mathbb{E}[f(\theta^{(t+1)})] \leq f(\theta^{(t)}) - \rho \|\nabla f(\theta^{(t)})\|^2, \quad (18)$$

where $\rho > 0$ depends on smoothness, client count, dissimilarity, and proximal strength. After $T = \mathcal{O}(\Delta/(\rho\epsilon))$ rounds, FedProx reaches an ϵ stationary point:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta^{(t)})\|^2] \leq \epsilon, \quad (19)$$

where $\Delta = f(\theta^{(0)}) - f^*$ is initial suboptimality.

Extension to partitioned parameters. In our framework, the global parameter set is partitioned into shared backbone and modality-specific block sets. The server updates the backbone using all clients (N_t), whereas each modality block averages only over same-modality clients ($N_t^{(m)}$). This ensures that each parameter block receives an *unbiased gradient estimate* of its corresponding sub-objective, allowing the standard FedProx analysis to be applied without modification. We do not claim a new convergence theorem; rather, we show that our aggregation scheme preserves the conditions under which FedProx guarantees hold. Empirically, this enables stable convergence despite severe non-IID skew ($\alpha = 0.3$), as validated in Section 4.

3.4.4. Computational and communication complexity

Per-round complexity. For client k modalities m and n_k samples, the local training cost per epoch is dominated by the Mamba backbone: $\mathcal{O}(n_k L_m d_{\text{model}} d_{\text{state}})$, which scales linearly in L_m (compared with the quadratic $\mathcal{O}(L_m^2)$ complexity of the transformer self-attention). The per-client communication cost equals the model update size: $\mathcal{O}(|\theta_{\text{in},m}| + |\theta_{\text{back}}| + |\theta_{\text{cls},m}|) \approx 1.70\text{--}1.85$ MB in our configuration (Section 4.5).

Algorithm 1 presents the complete federated training protocol integrating FedProx, modality-aware aggregation, and secure aggregation (formalized in Section 3.5).

3.5. Privacy-preserving secure aggregation

A core requirement of FedMamba is to protect client privacy from an honest-but-curious server. We achieve this through a design that is fully compatible with cryptographic secure aggregation protocols, specifically the protocol of Bonawitz et al. [5]. This section details our simulation approach, protocol compatibility, and formal integration with modality-aware aggregation.

3.5.1. Simulation rationale

In this study, we focused on the algorithmic convergence and robustness dynamics of FedMamba. To isolate these factors from system-level complexities and cryptographic implementation overhead, we *simulate* the secure aggregation protocol in a numerically equivalent manner as follows:

- 1) Each client pre-scales its update by local sample count n_k *before* any masking would occur.
- 2) The server securely computes the sum of scaled updates, recovering a single aggregate from which individual contributions cannot be inferred.
- 3) The server completes weighted averaging by partitioning this aggregate and dividing by public total counts N_t and $N_t^{(m)}$.

The simulation does not provide cryptographic privacy guarantees; it is standard practice in federated learning to isolate learning dynamics for reproducibility [3,4]. Real-world deployment requires integration with production cryptographic libraries (e.g., TensorFlow Federated and PySyft), and the performance implications are discussed in Section 4.5.3.

3.5.2. Compatibility with standard secure aggregation protocols

Our modality-aware aggregation treats secure aggregation as a black-box primitive and does not require changes to the underlying cryptographic protocol.

Client-side pre-scaling by n_k occurs before the masking. After protocol unmasking, the server recovers only the aggregate $\sum_{k \in S_t} n_k \Delta_k^{(t)}$, and never individual updates. Final weighted averaging divides the backbone components by N_t and modality-specific components by $N_t^{(m)}$, and operates on plaintext aggregates using public counts. Thus, the modality-aware weighting is achieved through:

- **Client side:** Multiply update by n_k (scalar operation, negligible cost).
- **Server side:** Partition unmasked aggregate and divide by public counts N_t , $N_t^{(m)}$ (no cryptographic operations).

This design preserves compatibility with any secure aggregation protocol that computes $\sum_k f(x_k)$ where f is client-side preprocessing (in our case, $f(\Delta_k) = n_k \Delta_k$).

3.5.3. Secure aggregation protocol formalization

We formalize our integration with the protocol of Bonawitz et al. [5]. Each client k generates cryptographic masks to protect its sample-weighted update $\Delta_{k,\text{scaled}}^{(t)} = n_k \Delta_k^{(t)}$:

- **Pairwise masks:** For each other client $j \neq k$, generate shared pseudorandom vector $\mathbf{R}_{k,j}^{(t)} \in \mathbb{R}^{|\theta|}$ from a common seed.
- **Private mask:** Generate $\mathbf{S}_k^{(t)} \in \mathbb{R}^{|\theta|}$ known only to client k .

Pairwise masks are cancelled when summed (client k adds $\mathbf{R}_{j,k}$ while client j subtracts $\mathbf{R}_{j,k}$), and private masks sum to a value the server can remove only in aggregate. Thus, the server never observes any individual client's update in cleartext.

The masked update transmitted by client k is:

$$\tilde{\Delta}_k^{(t)} = \Delta_{k,\text{scaled}}^{(t)} + \mathbf{S}_k^{(t)} + \sum_{j \in S_t, j < k} \mathbf{R}_{j,k}^{(t)} - \sum_{j \in S_t, j > k} \mathbf{R}_{k,j}^{(t)}. \quad (20)$$

After aggregating all masked updates, the server subtracts $\sum_{k \in S_t} \mathbf{S}_k^{(t)}$ (obtained via a separate secure protocol [5]) to recover:

$$\Delta_{\text{sum}}^{(t)} = \sum_{k \in S_t} n_k \Delta_k^{(t)}. \quad (21)$$

The server then applies modality-aware divisions (Eqs. (15)–(17)) to obtain final updates. This completes the integration of secure aggregation with our multimodal federated protocol, which is formalized in Algorithm 1.

Sample counts n_k are treated as public metadata, which is a standard assumption in federated learning [3]. For stricter privacy, counts can be bucketed (e.g., reporting ranges 100–200 instead of exact values) with minimal impact on weighted averaging accuracy.

Algorithm 1 FedMamba: Multimodal federated learning (MMFL) with FedProx regularization and modality-aware secure aggregation.

Require: Client data $\{D_k\}$, total rounds T , local epochs E , prox param. μ , participation ratio ρ

```

1: Server: Initialize global model  $\theta^{(0)}$ 
2: for  $t = 0$  to  $T - 1$  do
3:    $S_t \leftarrow$  Sample a fraction  $\rho$  of clients from each modality
4:   Broadcast  $\theta^{(t)}$  to all clients  $k \in S_t$ 
5:   Client  $k \in S_t$  (modality  $m_k$ ), in parallel:
6:      $\theta_k \leftarrow \theta^{(t)}$ 
7:     for  $e = 1$  to  $E$  do
8:       for minibatch  $B \subset D_k$  do
9:         Minimize  $\mathcal{L}_k(\theta_k; B) + \frac{\mu}{2} \|\theta_k - \theta^{(t)}\|_2^2$  by one GD step
10:      end for
11:    end for
12:     $n_k \leftarrow |D_k|$ 
13:     $\Delta_k^{(t)} \leftarrow \theta_k - \theta^{(t)}$  ▷ Raw model update
14:     $\Delta_{k,\text{scaled}}^{(t)} \leftarrow n_k \cdot \Delta_k^{(t)}$  ▷ Pre-scale for weighted secure agg
15:     $\tilde{\Delta}_k^{(t)} \leftarrow \text{Mask}(\Delta_{k,\text{scaled}}^{(t)}, k, t)$  ▷ Per Eq. (20)
16:    Send  $(\tilde{\Delta}_k^{(t)}, n_k, m_k)$  to server
17:  Server:
18:  Receive  $(\tilde{\Delta}_k^{(t)}, n_k, m_k)$  from all  $k \in S_t$ 
19:   $\tilde{\Delta}_{\text{sum}}^{(t)} \leftarrow \sum_{k \in S_t} \tilde{\Delta}_k^{(t)}$ 
20:   $\Delta_{\text{sum}}^{(t)} \leftarrow \text{Unmask}(\tilde{\Delta}_{\text{sum}}^{(t)})$  ▷  $\sum_k n_k \Delta_k^{(t)}$ 
21:  Compute  $N_t \leftarrow \sum_{k \in S_t} n_k$  and  $N_t^{(m)} \leftarrow \sum_{k \in S_t^{(m)}} n_k$  for each  $m \in \mathcal{M}$ 
22:  Partition  $\Delta_{\text{sum}}^{(t)}$  into:
23:     $\{\Delta_{\text{sum,back}}^{(t)}\}$ 
24:     $\{\Delta_{\text{sum,in},m}^{(t)}\}_{m \in \mathcal{M}}$ 
25:     $\{\Delta_{\text{sum,cls},m}^{(t)}\}_{m \in \mathcal{M}}$ 
26:   $\Delta_{\text{back}}^{(t+1)} \leftarrow \Delta_{\text{sum,back}}^{(t)} / N_t$ 
27:   $\Delta_{\text{in},m}^{(t+1)} \leftarrow \Delta_{\text{sum,in},m}^{(t)} / N_t^{(m)}, \quad \forall m \in \mathcal{M}$ 
28:   $\Delta_{\text{cls},m}^{(t+1)} \leftarrow \Delta_{\text{sum,cls},m}^{(t)} / N_t^{(m)}, \quad \forall m \in \mathcal{M}$ 
29:  Assemble  $\Delta^{(t+1)}$  from the parts
30:   $\theta^{(t+1)} \leftarrow \theta^{(t)} + \Delta^{(t+1)}$ 
31: end for
Ensure: Global model  $\theta^{(T)}$ 

```

3.6. Hardening the framework with federated adversarial training

To improve resilience against inference-time evasion attacks, we integrated a federated adversarial training (FAT) scheme into the framework. The objective of FAT extended beyond hardening individual clients; it leveraged the federated learning process to collaboratively construct a single and globally robust model. While adversarial examples were generated on local client data, defensive training enhanced feature representations within the shared Mamba backbone (θ_{back}). When clients transmitted their model updates, these robustness-enhancing gradients were aggregated by the server. This process effectively disseminated the learned resilience across the entire network, allowing clients to collectively contribute to a global defense that was stronger than any client could build independently. This complemented the FedProx regularization used to ensure stable convergence [21,26].

3.6.1. Modality-specific adversarial attack generation

We generated adversarial perturbations for clients designated for robust training (HDFS and SWaT) using Projected Gradient Descent (PGD) [21], an iterative method for worst-case examples.

We explicitly trained on the hardest nearby inputs, so the classifier learned margins that were robust to small, plausible perturbations. Constraining perturbations to an ℓ_∞ ball kept attacks realistic for each feature space. For a given client k with local parameters θ_k , PGD sought a perturbed input x' that maximized the single-sample loss ℓ within an ℓ_∞ ball of radius ϵ around the clean input x :

$$x^* = \arg \max_{x' \in B_\infty(x, \epsilon)} \ell(f(x'; \theta_k), y), \quad (22)$$

where $B_\infty(x, \epsilon) = \{x' : \|x' - x\|_\infty \leq \epsilon\}$.

This maximization is approximated over K_{PGD} iterations, starting from $x^{(0)} = x$:

$$x^{(i+1)} = \Pi_{\mathcal{B}_{\infty}(x, \epsilon)} \left(x^{(i)} + \alpha_{\text{PGD}} \cdot \text{sgn} \left(\nabla_{x^{(i)}} \ell(f(x^{(i)}; \theta_k), y) \right) \right),$$

$$i = 0, \dots, K_{\text{PGD}} - 1. \quad (23)$$

where α_{PGD} was the step size and $\Pi_{\mathcal{B}_{\infty}(x, \epsilon)}$ denoted projection onto $\mathcal{B}_{\infty}(x, \epsilon)$. We set $x' = x^{(K_{\text{PGD}})}$. The attack parameters ($\epsilon, \alpha_{\text{PGD}}, K_{\text{PGD}}$) were configured per modality, and we implemented the PGD using the Adversarial Robustness Toolbox (ART) [27].

3.6.1.1. Attack on continuous data (SWaT and CICIoMT-2024). For clients with continuous tabular data, the PGD attack was directly applied to the input features. The projection operator Π in Eq. (23) is crucial as it enforces both the ℓ_{∞} constraint and valid feature ranges (i.e., `clip_values`), ensuring that adversarial examples remain physically or logically plausible.

3.6.1.2. Attack on discrete data (HDFS). For HDFS clients, perturbing discrete input tokens is ill-defined. Instead, we applied the attack in the continuous embedding space, which is standard. The input sequence $x^{(\text{H})}$ was first passed through the embedding layer to obtain $\mathbf{H}_0^{(\text{H})}$, and PGD was then performed on $\mathbf{H}_0^{(\text{H})}$ to generate a perturbed embedding that was subsequently passed to a shared backbone.

3.6.2. Robust local objective function

An initial robustness evaluation of our standard federated model revealed that while the HDFS and SWaT modalities are highly susceptible to PGD attacks, the CICIoMT-2024 modality is inherently robust. This is likely because the high-dimensional continuous feature spaces of HDFS (in its embedding space) and SWaT are more vulnerable to small adversarial perturbations than the CICIoMT-2024 statistical network flow features. Based on this finding, we selectively applied adversarial training: clients for HDFS and SWaT were trained using a robust objective, whereas the CICIoMT-2024 clients continued with standard training. For each mini-batch (x, y) , an adversarial example x' is generated using the PGD method described above. For a minibatch $B \subset \mathcal{D}_k$, the local model θ_k was updated by minimizing the following robust objective function $\mathcal{J}_k(\theta_k)$:

$$\mathcal{J}_k(\theta_k) = \frac{1}{|B|} \sum_{(x,y) \in B} \left[\ell(f(x; \theta_k), y) + \ell(f(x'; \theta_k), y) \right] + \frac{\mu}{2} \|\theta_k - \theta^{(t)}\|_2^2. \quad (24)$$

Clients not designated for robust training (CICIoMT-2024) minimized the standard FedProx objective (Eq. (14)) without an adversarial term. This hybrid approach allowed the federated system to learn robust features in the shared backbone θ_{back} from the contributions of the HDFS and SWaT clients, whereas the CICIoMT-2024 clients focused on the standard-task performance. After local training, all clients transmitted their updates to the server for modality-aware secure aggregation (Algorithm 2).

Algorithm 2 Robust local training with projected gradient descent adversarial training (PGD-AT) and FedProx.

Require: Global model $\theta^{(t)}$, local dataset $\mathcal{D}_k$, modality m_k , epochs E , prox param. μ , PGD params ($\epsilon, \alpha_{\text{PGD}}, K_{\text{PGD}}$)

```

1:  $\theta_k \leftarrow \theta^{(t)}$ 
2: for  $e = 1$  to  $E$  do
3:   for minibatch  $B \subset \mathcal{D}_k$  do
4:     Initialize batch loss  $\mathcal{L}_B \leftarrow 0$ 
5:     for each  $(x, y) \in B$  do
6:       if  $m_k \in \{\text{HDFS, SWaT}\}$  then ▷ Selective adversarial training
7:         Generate  $x'$  from  $x$  by applying  $K_{\text{PGD}}$  steps of PGD (Eq. (23))
8:          $\mathcal{L}_B \leftarrow \mathcal{L}_B + \ell(f(x; \theta_k), y) + \ell(f(x'; \theta_k), y)$ 
9:       else ▷ Standard training for CICIoMT-2024
10:         $\mathcal{L}_B \leftarrow \mathcal{L}_B + \ell(f(x; \theta_k), y)$ 
11:      end if
12:    end for
13:     $\mathcal{J}_k \leftarrow \frac{1}{|B|} \mathcal{L}_B + \frac{\mu}{2} \|\theta_k - \theta^{(t)}\|_2^2$  ▷ Eq. (24)
14:    Update  $\theta_k$  with one step of gradient descent on  $\mathcal{J}_k$ 
15:  end for
16: end for return  $\theta_k$ 

```

4. Experimental evaluation and results

In this section, we present a comprehensive empirical evaluation of the FedMamba framework. Our analysis is structured to first validate the performance of our federated approach against centralized baselines using the clean data. We then diagnosed the inherent vulnerabilities of the model to evasion attacks and subsequently demonstrated the effectiveness of our proposed federated adversarial training defense. Finally, an ablation study validated the choice of the FedProx algorithm.

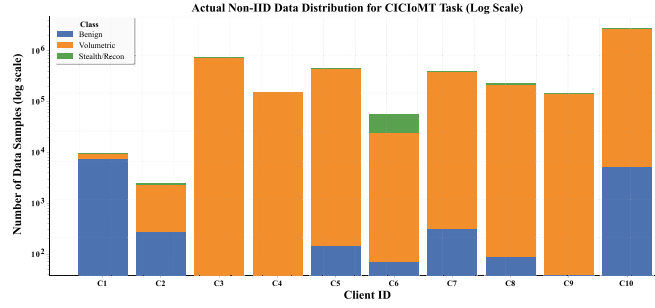


Fig. 3. A representative non-IID data distribution for CICIoMT-2024, visualized for a subset of 10 clients, generated via a Dirichlet distribution ($\alpha = 0.3$). The plot demonstrates the severe statistical heterogeneity that clients must overcome, where each client holds a highly skewed and incomplete subset of the global class distribution. The y-axis is presented on a logarithmic scale to visualize the distribution across clients with vastly different sample sizes. This visualization illustrates the severe statistical heterogeneity induced by the Dirichlet partitioning; its impact on FedAvg and the motivation for FedProx are quantified later in Table 4.

4.1. Experimental setup

4.1.1. Datasets and preprocessing

Each dataset was split into training, validation, and test partitions with proportions 70%, 15%, and 15%. For binary tasks, we selected a decision threshold τ on the validation set by maximizing the macro F1-score over a uniform grid on $[0.05, 0.95]$ with a step 0.05. For the multiclass CICIoMT-2024 task, we use the $\arg \max$ class prediction (no decision threshold). We report the Macro F1-score for the held-out test sets for all modalities.

4.1.2. Federated environment simulation

We simulated statistical heterogeneity using a Dirichlet sampler with a concentration $\alpha = 0.3$ per class to distribute indices across clients, followed by a non-empty enforcement pass. The resulting severe data skew for the CICIoMT-2024 task is shown in Fig. 3. In the baseline configuration, we use $K = 35$ clients in total (distributed across modalities). In each round, we sampled a fraction $\rho = 0.4$ of clients per modality uniformly at random to form S_r . For clarity, Fig. 3 visualizes the CICIoMT-2024 non-IID split for a representative subset of 10 clients.

4.2. Experimental baselines and evaluation metrics

To rigorously evaluate our framework, we established a clear set of baselines and metrics. We evaluated our framework against the following benchmarks:

- 1) Centralized Specialists: Individual Mamba models trained centrally on each dataset, representing a non-federated performance upper bound.
- 2) FedMamba (Standard, FedProx): Our primary federated model, trained using FedProx and secure aggregation on non-IID data.
- 3) FedMamba-AT (Robust): Our final proposed model, hardened using the selective federated adversarial training scheme from Section 3.6.
- 4) Weighted FedAvg: For the ablation study, we compare standard Federated Averaging under both IID and non-IID data partitions to isolate the impact of FedProx.

We evaluated the Accuracy, Precision, Recall, and Macro F1-score of the proposed framework. Owing to the class imbalance, the Macro F1-score was the primary metric. For binary tasks, we also reported the false-positive rate (FPR) and ROC-AUC. For CICIoMT-2024, we report one-vs-rest macro-averaged ROC-AUC. The adversarial evaluation settings are summarized in Table B.3.

4.3. Performance and robustness analysis

We next present our empirical results to demonstrate the efficacy and resilience of the proposed framework.

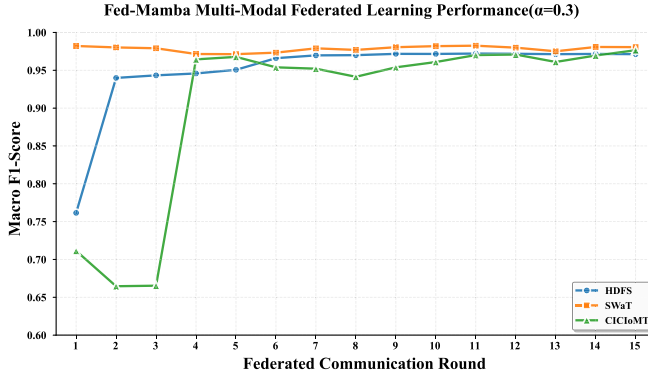
4.3.1. Baseline performance of the federated framework

Our first evaluation established the performance of the standard FedMamba model (trained with FedProx) on clean, unseen test data and compared it with that of the centralized specialist upper bound. The results in Table 2 show that our federated approach achieved near-centralized performance for all modalities.

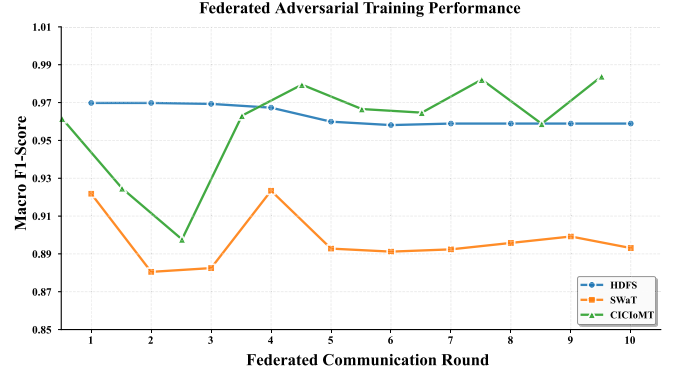
The performance gap between the federated and centralized models was minimal. As shown in Fig. 4a, the standard FedMamba model achieved stable convergence across all modalities, confirming that the benefits of privacy-preserving decentralized training were achieved with little to no trade-off in accuracy on clean data. Notably, the federated model's performance on the complex CICIoMT-2024 task was superior to its siloed counterpart, suggesting that the shared backbone facilitated positive knowledge transfer, acting as a powerful regularizer.

Table 2
Centralized Models vs. Standard FedMamba on clean data.

Model / task	Macro F1-score	Accuracy	ROC-AUC
Centralized HDFS	0.9900	0.9986	0.9975
FedMamba (HDFS)	0.9584	0.9980	0.9751
Centralized SWaT	0.9852	0.9936	0.9998
FedMamba (SWaT)	0.9795	0.9957	0.9994
Centralized CICIoMT-2024	0.9495	0.9990	0.9999
FedMamba (CICIoMT-2024)	0.9665	0.9994	0.9999



(a) Convergence of Standard FedMamba on Clean Data



(b) Convergence of Robust FedMamba-AT with adversarial Data

Fig. 4. Convergence under non-IID partitioning ($\alpha=0.3$). (a) The standard FedMamba converges stably on clean data across modalities. (b) Robust FedMamba-AT shows the expected accuracy-robustness trade-off during adversarial training while maintaining high final performance, especially evident on SWaT.

Table 3

Macro F1 on clean, black-box Gaussian noise, transfer black-box, and white-box PGD. Budgets match our setup (noise std σ for Gaussian; ℓ_∞ radius ϵ for PGD and transfer). Transfer uses PGD crafted on the alternate model and evaluated zero-query on the target.

Dataset	Attack	Budget	FedMamba	FedMamba-AT	Δ AT-Std
<i>Macro F1 (higher is better); Clean baselines shown for reference</i>					
HDFS	Clean	–	0.9584	0.9486	-0.0098
HDFS	Gaussian	$\sigma=0.01$	0.9617	0.9486	-0.0131
HDFS	Transfer	$\epsilon=0.01$	0.3800	0.9478	+0.5678
HDFS	PGD (WB)	$\epsilon=0.01$	0.3791	0.9480	+0.5689
SWaT	Clean	–	0.9795	0.8578	-0.1217
SWaT	Gaussian	$\sigma=0.05$	0.9737	0.8598	-0.1139
SWaT	Transfer	$\epsilon=0.05$	0.7161	0.8059	+0.0898
SWaT	PGD (WB)	$\epsilon=0.05$	0.5147	0.8357	+0.3210
CICIoMT-2024	Clean	–	0.9665	0.9741	+0.0076
CICIoMT-2024	Gaussian	$\sigma=0.05$	0.9657	0.9733	+0.0076
CICIoMT-2024	Transfer	$\epsilon=0.05$	0.9598	0.9737	+0.0139
CICIoMT-2024	PGD (WB)	$\epsilon=0.05$	0.9665	0.9645	-0.0020

4.3.2. Adversarial vulnerability of the standard model

We first assessed the baseline robustness of the standard FedMamba. Table 3 reported Macro F1-score on clean data as well as black-box Gaussian noise and transfer-based attacks. The Auto-PGD [28] white-box results are presented in Fig. 5. As shown there, the standard model degraded sharply on the HDFS and SWaT datasets, whereas CICIoMT-2024 remained relatively stable.

The results revealed significant vulnerabilities. Although the CICIoMT-2024 task remained largely unaffected, the model's performance on HDFS and SWaT collapsed under attack. The Macro F1-scores plummeted by **60.4%** for HDFS and **47.5%** for SWaT. This degradation underscored that high performance on clean data offered no guarantee of security and provided a strong motivation for our proposed defensive mechanism.

4.3.3. Effectiveness of federated adversarial training

We evaluated the efficacy of our final robust model, FedMamba-AT. Table 3 compared clean and PGD performance side by side with the standard model.

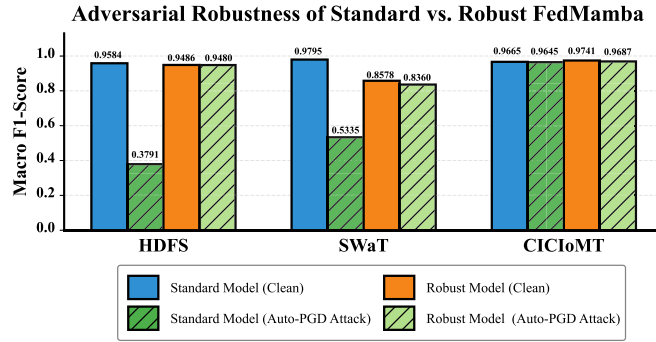


Fig. 5. Adversarial robustness under Auto-PGD. The bars show the macro F1 for the standard FedMamba and robust FedMamba-AT on clean data and under an Auto-PGD white-box attack with an ℓ_∞ constraint. Attack budgets and loss settings follow Table B.3 and Section 3.6.

Table 4

Ablation: aggregation under IID and non-IID on clean data.

Task	Model /Distribution	Macro F1-score	Acc. (%)	ROC AUC
HDFS	Weighted FedAvg (IID)	0.9844	0.9992	0.9971
	Weighted FedAvg (non-IID)	0.9840	0.9990	0.9971
	FedProx (non-IID)	0.9584	0.9980	0.9751
SWaT	Weighted FedAvg (IID)	0.9834	0.9965	0.9992
	Weighted FedAvg (non-IID)	0.9734	0.9965	0.9990
	FedProx (non-IID)	0.9795	0.9957	0.9994
CICIoMT-2024	Weighted FEDAVG (IID)	0.9550	0.9991	0.9999
	Weighted FEDAVG (non-IID)	0.8550	0.9591	0.8050
	FedProx (non-IID)	0.9665	0.9994	0.9999

The results in Table 3 confirm the success of our collaboratively hardened strategy. For the HDFS task, the model was exceptionally resilient, with its Macro F1-score under the PGD attack recovering from 0.3791 to a robust **0.9480**. Similarly, the SWaT task’s robust Macro F1-score of **0.8357** indicates a dramatic improvement over the vulnerable model’s score of 0.5147. This gain introduced a modest accuracy-robustness trade-off, as the model learned a smoother decision boundary (Fig. 4b). Fig. 5 shows the Auto-PGD for stronger white-box stress testing, whereas Table 3 shows the standard PGD (WB) protocol for comparability with prior work.

This outcome demonstrates that adversarial training performed locally by HDFS and SWaT clients successfully encoded robust features into the shared Mamba backbone. Through federated aggregation, this learned resilience was integrated into the global model and disseminated to all clients, creating a collective defense that was stronger than isolated training. Crucially, the CICIoMT-2024 task, which did not perform local adversarial training, maintained a high performance, indicating that the robustness-enhancing updates did not cause negative interference. This validated the stability of our selective and collaborative defense (Fig. 5).

4.4. Ablation study

To validate the core design choices of our framework, we conducted a series of ablation studies focusing on the aggregation protocol, shared backbone architecture, and sequence model.

To justify our choice of FedProx over standard FedAvg in a non-IID environment, we performed an ablation. As shown in Table 4, Weighted FedAvg performed well under ideal IID splits, but under realistic non-IID skew, its performance degraded on CICIoMT-2024 while remaining competitive on HDFS and SWaT. In contrast, FedProx mitigated client drift and restored stable convergence on CICIoMT-2024 and across all modalities.

This study revealed the critical impact of statistical heterogeneity and validated our choice of FedProx. While Weighted FedAvg performed well under ideal IID conditions, its performance on the complex, multiclass CICIoMT-2024 task collapsed under the realistic non-IID distribution, with the Macro F1-score dropping from 0.9550 to 0.8550. This was a clear manifestation of client drift, exacerbated by the extreme data skew, as shown in Fig. 3, where clients such as C3 and C4 possessed zero samples of the “Benign” class. Unconstrained local training on highly biased data leads to diverged model updates that cannot be effectively averaged, causing performance collapse. In contrast, under the same non-IID conditions, FedProx prevented this collapse. Its proximal term mitigated client drift, enabling stable convergence and achieving a final F1-score of 0.9665, which even surpassed the IID baseline. This result validates FedProx as a necessary component for stable convergence in heterogeneous multimodal federated environments.

4.4.1. The impact of the shared backbone

To quantify the benefits of our multimodal architecture and directly address the advantages of cross-modal knowledge transfer, we conducted an ablation study. We compared our unified FedMamba model with a “No-Share” baseline, which was functionally

Table 5

Ablation study: Performance and efficiency of unified vs. no-share architectures under non-IID conditions.

Task	FedMamba (Unified)	No-Share Baseline	No-Share Baseline (Tuned)
HDFS	0.9713	0.9626	0.9821
SWaT	0.9806	0.5850	0.9448
CICIoMT-2024	0.9665	0.9629	0.9748
Total Parameters	0.54 M		1.41 M
Total Model Size	2.04 MB		5.39 MB

Table 6

Architectural ablation: FedMamba vs. FedTransformer (performance and efficiency metrics).

Metric	FedMamba	FedTransformer
<i>Performance (Macro F1-score)</i>		
HDFS	0.9584	0.9977
SWaT	0.9795	0.9770
CICIoMT-2024	0.9665	0.9660
<i>Efficiency</i>		
Total Parameters	0.54 M	1.18 M
Comm. Cost / Client (Avg)	~1.8 MB	~4.2 MB
Inference Latency / Sample	~0.012 ms	~0.011 ms

Table 7

Scalability of FedMamba with increasing client count.

Client Scale	Total Clients	Final Macro F1-score			Rounds to Converge (Approx.)
		HDFS	SWaT	CICIoMT2024	
1x	35	0.9584	0.9795	0.9665	~12
2x	70	0.9420	0.9750	0.9469	~11
3x	105	0.9414	0.9804	0.9716	~12

equivalent to three independent federated learning systems running in parallel with separate, unshared Mamba backbones for each modality. Both architectures were subjected to the same severe non-IID client distribution ($\alpha = 0.3$) and FedProx protocol.

The results in Table 5 show that the unified FedMamba model with a shared backbone outperformed all tasks. Most notably, the performance on the SWaT task collapsed in the standard no-share model, which did not converge effectively. A detailed analysis of the client data revealed the cause: the SWaT distribution was polarized, with clients split into “Normal”-only and “Attack”-only factions, leading to conflicting gradients that destabilized the isolated model. To provide a stronger baseline, we conducted a second experiment in which the No-Share SWaT model was given advantages: warm-starting from a pre-trained checkpoint and using a less restrictive FedProx regularization ($\mu = 0.001$). Even with this assistance, the tuned No-Share model’s performance (0.9448 F1-score) remained below that of the unified model (0.9806 F1-score).

4.4.2. Architectural choice: Mamba vs. transformer

To further validate our selection of Mamba as the backbone, we conducted an architectural ablation by comparing FedMamba with an equivalent framework using a standard transformer backbone, FedTransformer. Both models were trained using the same federated protocols. The results in Table 6 reveal a trade-off between the model performance and resource efficiency.

The FedTransformer model achieved near-perfect accuracy in the HDFS log analysis task (0.9977 F1-score), outperforming FedMamba. However, this came at a significant cost: the transformer-based model was more than 2× larger (1.18 M vs. 0.54 M parameters) and required more than 2× the communication bandwidth per client upload (~4.2 MB vs. ~1.8 MB).

This efficiency gap is a major consideration for IIoT security, where models are often deployed on edge devices with limited memory, power, and network bandwidth. FedMamba delivered strong, reliable performance across all three modalities while being substantially more lightweight and communication-efficient. Given that its inference latency is comparably low, FedMamba is a more practical and scalable solution for real-world deployment. Thus, this study confirmed that Mamba is a compelling architectural choice that strikes an excellent balance between high accuracy and the operational constraints of the IoT ecosystem. Table 6 quantifies this trade-off. Fig. 6 shows that the convergence trends matched the Mamba curves despite the larger model and upload.

4.5. Scalability and efficiency analysis

Beyond its accuracy, the practical feasibility of FedMamba was assessed by analyzing its scalability with an increasing number of clients and its overall computational efficiency.

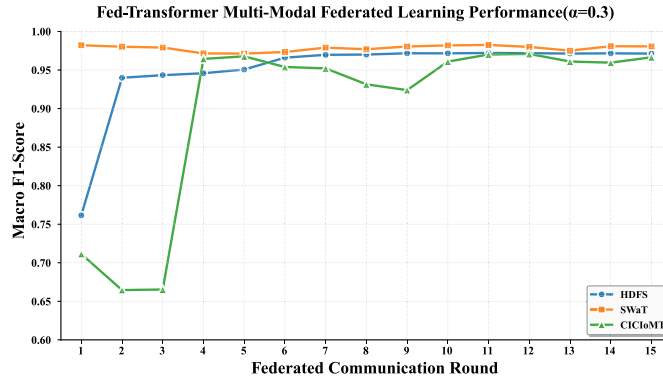


Fig. 6. FedTransformer convergence under non-IID partition ($\alpha=0.3$). Macro F1 vs. round on HDFS, SWaT, and CICIoMT-2024. Trends mirror Fig. 4a while requiring $2\times$ parameters and upload (Table 6).

Table 8

Projected upload time with secure aggregation (SA). The plaintext payload uses the mid-point 1.80 MB; the SA expansion factor uses a representative $1.73\times$ consistent with [5].

Condition	Payload	10 Mbps	100 Mbps
Plaintext	1.80 MB	1.44 s	0.144 s
With SA	3.11 MB	2.49 s	0.249 s
Overhead	+ 1.31 MB	+ 1.05 s	+ 0.105 s

4.5.1. Performance under scale

We conducted experiments by scaling the number of clients to $2\times$ (70 total) and $3\times$ (105 total) of the baseline configuration. The objective was to determine whether the model accuracy or convergence speed degraded as the client pool increased. The results, summarized in Table 7, demonstrate the robustness and scalability of the framework. Across all scales, the final Macro F1-scores remained remarkably stable, with the SWaT task achieving 0.9795 with 35 clients and 0.9804 with 105 clients. Furthermore, the model consistently reached peak performance within approximately 11–12 rounds, confirming that FedMamba can be scaled to larger client populations without sacrificing its performance and efficiency.

4.5.2. Efficiency and cost analysis

The unified architecture of FedMamba provides significant efficiency gains, as validated by our ablation study in Table 5. The unified model contained only **0.54M** parameters, a **62.1% reduction** compared with the 1.41M parameters required by the no-share baseline. This smaller footprint directly translated to lower communication overhead; a client upload was a modest 1.70–1.85 MB, and the total global model size was only 2.04 MB, compared with 5.39 MB for the three separate models in the No-Share baseline.

Computationally, the linear complexity of the Mamba backbone, $\mathcal{O}(L_m)$, provides a distinct advantage over the quadratic complexity of transformer-based alternatives, particularly for long sequences in IIoT data. For real-time deployment, the converged model was highly performant, demonstrating an inference latency of only 0.012–0.013ms per sample on an NVIDIA RTX 3090 GPU. This high throughput confirms the suitability of FedMamba for efficient and real-time threat detection in high-velocity data environments.

4.5.3. Secure aggregation: Deployment feasibility and overhead

We simulated secure aggregation (SA) to isolate the learning dynamics. Here, we summarize the feasibility and expected overhead when deploying a production SA protocol.

Protocol compatibility. Our modality-aware partitioned averaging remains a black-box consumer of SA in the sense of Bonawitz et al. [5]. Client-side pre-scaling by n_k occurs before masking, the server recovers only the masked sum, and the final divisions by N_t and $N_t^{(m)}$ occur on the unmasked aggregate. No changes to the cryptographic primitives are required, which keeps the design compatible with common SA implementations.

Expected overhead. Prior reports for SA rounds with dropout resilience indicate a communication expansion in the range of $1.6\text{--}1.8\times$ for the MB-scale model updates. For our update size (0.54M parameters, 1.70–1.85 MB), this yields a masked upload of roughly 2.9–3.3 MB per client. Table 8 instantiates a representative mid-point factor to show the end-to-end upload time impact at two link speeds.

Client-side masking cost scales linearly with the update length and is typically on the order of seconds on edge-class hardware for MB-scale updates, while server-side unmasking adds a small fraction of the aggregation time. Our pre-scaling and partitioned divisions are algebraic operations on the unmasked sum and are negligible compared to the network transfer.

Table 9
Comparative analysis of state-of-the-art IoT/ICS security frameworks.

Framework	Core Methodology	Key Reported Result	Multi-Modal	Feder-ated	Adversarially Robust
Zeng et al. [14]	Federated Neural Architecture Search + AT	0.961 F1-score (SWaT)	✗	✓	✓
Kim [19]	Centralized RL-based Attack & Defense	0.80 Attack Success (SWaT)	✗	✗	✓ (Centralized)
Albanbay et al. [29]	Standard Federated Learning	0.87 F1-score (BoT-IoT)	✗	✓	✗
Chatterjee et al. [13]	Federated Hybrid Ensemble	0.93 F1-score (CICIDS2017)	✗	✓	✗
Rajesh et al. [15]	Federated Transfer Learning (Combo-NN)	0.974 F1-score (IIoT)	✗	✓	✗
FedMamba (Ours)	Multimodal Federated Mamba + AT	0.948-0.974 F1-score (Table 3)	✓	✓	✓

Production SA uses a dropout-resilient, round structure. Therefore, the round duration includes a synchronization term determined by the participation ratio ρ and timeout settings. In practice, this term is a small, fixed addition per round for the client scales we target.

If integer packing or fixed-point quantization is enabled by the SA stack, the plaintext update can be reduced before masking, which reduces the masked payload proportionally.

Deployment guidance. *Tier 1 (trusted server).* The SA can be omitted when the server is fully trusted. *Tier 2 (honest-but-curious, our threat model).* Use standard SA with dropout resilience; expect a round-time increase on the order of 10–15% for the update sizes and link speeds above. *Tier 3 (fault/Byzantine tolerance).* Robust SA variants can be layered where required; this is orthogonal to the learning rule.

The deployment of SA is feasible with a well-characterized cost. The dominant component is a predictable communication expansion, while the compute and orchestration terms are smaller. Our modality-aware aggregation introduces no additional cryptographic complexity beyond that of standard federated learning.

4.6. Comparative analysis with state-of-the-art frameworks

To contextualize the contributions of FedMamba, we conducted a comparative analysis with representative state-of-the-art intrusion detection systems designed for the IIoT and cyber-physical systems. Although direct numerical comparisons were precluded by differences in datasets and experimental conditions, the overview in Table 9 highlights how FedMamba addresses critical limitations in the literature.

The literature reveals a trend toward increasingly sophisticated FL-based detection frameworks. Methods such as those of Albanbay et al. [29] and Chatterjee et al. [13] demonstrate strong performance under unimodal federated contexts. Zeng et al. [14] introduced adversarial training (AT) within a federated framework for sensor data, and Rajesh et al. [15] utilized federated transfer learning to enhance generalization. Concurrently, Kim [19] underscored the severity of the adversarial threat landscape in ICS within a centralized paradigm.

This analysis revealed a persistent gap in the literature. Although prior studies have addressed these challenges in isolation, no framework has provided a holistic solution that jointly satisfies privacy-preserving decentralized learning (federated), multimodal processing, and resilience against evasion attacks. To the best of our knowledge, FedMamba is the first framework to systematically integrate and validate these three capabilities.

4.7. Limitations and future work

Our secure aggregation is simulated, so future work will integrate production cryptographic protocols and quantify end-to-end overhead in real deployments. The current evaluation focuses on structured IIoT modalities and three datasets, thus extending FedMamba to unstructured streams such as audio and video and to broader benchmarks remains open. We assume one modality per client and prioritize inference-time evasion, so relaxing this assumption and addressing stronger training-time threats such as poisoning, backdoors, and Byzantine failures is a priority. Our selective adversarial training covers only HDFS and SWaT with fixed ℓ_∞ budgets, so adaptive, budget-agnostic defenses and certified robustness are promising directions. Finally, resource-aware scheduling, client drift detection, and privacy auditing of update leakage can improve practicality at scale.

5. Conclusion

This study introduced FedMamba, an MMFL framework designed for heterogeneous IIoT and critical infrastructure environments. We address three challenges: pronounced data heterogeneity across modalities, statistically non-IID client distributions, and the joint requirements of privacy awareness and adversarial robustness. The contributions of this study are as follows: (i) a unified Mamba-based architecture that processes and integrates diverse security modalities; (ii) a robust, privacy-aware federated training protocol built on FedProx for convergence stability and a simulated secure aggregation layer compatible with practical protocols; and (iii) an integrated federated adversarial training scheme that hardens the model against inference-time evasion.

Experiments validated the effectiveness of our proposed approach. The standard FedMamba model performed well on clean data, achieving Macro F1-scores of 0.9584, 0.9795, and 0.9665 on HDFS, SWaT, and CICIoMT-2024, respectively, which are competitive with centralized baselines. Under strong PGD attacks, performance on HDFS and SWaT degraded significantly (to 0.3791 and 0.5147), indicating vulnerability, whereas CICIoMT-2024 remained inherently robust. The adversarially trained FedMamba-AT restored resilience, sustaining F1-scores of 0.9480 on HDFS and 0.8357 on SWaT. These results demonstrate that a multimodal, federated, and adversarially trained design can substantially improve robustness while preserving performance. Despite these strengths, FedMamba has limitations. First, our simulated secure aggregation omits the communication and computational overhead of real-world cryptography. Second, while the framework excels in structured IIoT data, its scalability to unstructured modalities (e.g., video) remains to be explored. Future work will focus on integrating production-ready cryptographic libraries, extending the framework to new data types, and broadening the threat model to include training-time attacks such as data poisoning.

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Data availability

SWaT and CICIoMT-2024 are publicly available; HDFS log data is publicly available (see References). Processed splits and training scripts will be provided in an online repository.

Code availability

An artifact containing the code and configuration files will be made available upon request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Hafiz Bilal Ahmad: Conceptualization, Methodology, Software, Writing - original draft; **Haichang Gao:** Supervision, Funding acquisition, Project administration, Writing - review & editing; **Naila Latif:** Visualization, Writing - review & editing; **Tanjeena Manzoor:** Visualization, Writing - review & editing.

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Appendix A. Notation reference

Appendix B. Reproducibility and implementation details

This section provides the minimal configuration required to reproduce our results. We first list the software environment, followed by hardware, data handling, training and federated configuration, and finally, the adversarial evaluation budgets.

B.1. Software and framework versions

The complete software stack used in all the experiments is summarized in [Table B.1](#).

B.2. Hardware

All experiments were conducted on a Linux x86_64 workstation with an NVIDIA RTX 3090 (24 GB VRAM), a 12-core desktop-class CPU, and 64 GB system RAM.

Table A.1

Complete notation used throughout the paper.

Symbol	Description
$\mathcal{M}$	Set of modalities (HDFS, SWaT, CICIoMT-2024)
$\mathcal{K}$	Set of clients
S_t	Subset of participating clients at round t
$S_t^{(m)}$	Participating clients of modality m at round t
θ	All model parameters
$\theta_{in,m}$	Input adapter parameters for modality m
θ_{back}	Shared backbone parameters
$\theta_{cls,m}$	Classifier head parameters for modality m
d_{model}	Shared latent dimension
d_{state}	State dimension of the Mamba model
L_m	Sequence length for modality m
C	Number of classes
B	Minibatch size
$\mathcal{B}$	A minibatch, $\mathcal{B} \subset \mathcal{D}_k$
n_k	Number of samples on client k
N_t	Total samples across S_t in round t
$N_t^{(m)}$	Total samples of modality m across S_t in round t
μ	FedProx proximal regularization parameter
α	Dirichlet concentration parameter for partitioning
ρ	Client participation ratio per round
τ	Validation-selected decision threshold (binary tasks)
ϵ	ℓ_∞ adversarial budget (PGD)
α_{PGD}	PGD step size
K_{PGD}	Number of PGD iterations
$\Pi_{\mathcal{B}_\infty(x,\epsilon)}$	Projection onto the ℓ_∞ ball of radius ϵ around x
$f_{in,m}(\cdot; \theta_{in,m})$	Modality- m input adapter
$f_{cls,m}(\cdot; \theta_{cls,m})$	Modality- m classifier head
$\text{Mamba}(\cdot; \theta_{back})$	Shared Mamba backbone
$\text{LN}(\cdot)$	LayerNorm operator
$\mathbf{h}_t^{(m)}$	Hidden state at time t for modality m
$\mathbf{z}^{(m)}$	Pooled representation for modality m (last-step + LN)

Table B.1

Software stack used in all experiments.

Component	Version	Notes
Python	3.10	64-bit
PyTorch	2.4.0	CUDA runtime 11.8
CUDA	11.8	For PyTorch 2.4.0
mamba_ssm	2.2.5	cu11, torch2.4, cxx11abiFALSE
causal_conv1d	1.5.2	cu11, torch2.4, cxx11abiTRUE
Adversarial Robustness Toolbox	1.16.0	Evaluation and PGD
NumPy	1.26.x	
pandas	2.2.x	
scikit-learn	1.5.x	
openpyxl	3.1.x	SWaT loader

B.3. Data handling

Unless noted in [Section 4.1](#), we used the following preprocessing across all runs: HDFS used tokenized log sequences of length 50 with left padding; SWaT used sliding windows of length 60 with standardization fitted on the training split; and CICIoMT-2024 was treated as a single-step tabular input with standardization fitted on the training split. The dataset splits were 70/15/15 for training/validation/testing. For HDFS and SWaT, the decision threshold was selected in the validation set by a macro-F1 grid search. Label maps are HDFS Normal, Anomaly→0,1, SWaT Normal, Attack→0,1, and CICIoMT-2024 Normal, DoS, Other→0,1,2.

B.4. Training and federated configuration

The centralized hyperparameters and FL protocol are listed in [Table B.2](#). We implemented the model in PyTorch with `mamba_ssm`; we used AdamW, differential learning rates, FedProx for non-IID stability, sample-weighted block-wise aggregation, and simulated secure aggregation via client pre-scaling by n_k .

Table B.2
Unified training and federated configuration.

Item	Value
Backbone	Mamba, $d_{\text{model}} = 256$, $d_{\text{state}} = 16$, $d_{\text{conv}} = 4$, expand 2
Pooling	last token + LayerNorm
Heads	HDFS binary, SWaT binary, CICIoMT-2024 3-class
Optimizer	AdamW; weight decay 1×10^{-2}
LR shared backbone	5×10^{-6}
LR adapters + heads	1×10^{-4}
Batch size	64; grad clip global L2 = 1.0
Rounds T	15; local epochs $E=2$; client fraction $\rho=0.4$
Non-IID	Dirichlet $\alpha=0.3$ per class
FedProx	$\mu=0.01$
Aggregation	sample-weighted backbone over all; modality blocks within modality
Secure aggregation	simulated; client pre-scaling by n_k

Table B.3
Adversarial evaluation settings.

Task	Space	Norm	ϵ	Step	Iter.
HDFS	Embedding	L_{∞}	0.01	0.002	10
SWaT	Input	L_{∞}	0.05	0.01	10
CICIoMT-2024	Input	L_{∞}	0.05	0.01	10

Table D.1
Backdoor probe: clean metrics and ASR for the standard FedMamba model.

Dataset	Macro F1	Acc.	ASR
HDFS	0.9718	0.9986	0.0000
SWaT	0.9741	0.9947	0.0061
CICIoMT-2024	0.9098	0.9988	0.0619

B.5. Attack protocols and budgets

The adversarial evaluation settings are summarized in Table B.3. The transfer black box uses the same budgets crafted on a surrogate and evaluates zero queries on the target. Gaussian noise uses the standard deviations listed.

B.6. Data and code availability

Preprocessing scripts, configuration files (including seeds), and FedMamba training code will be released upon acceptance at an anonymized repository. Where licensing permits, we will also provide the train/val/test index splits for HDFS, SWaT, and CICIoMT-2024 to facilitate exact replication.

C. Secure aggregation microbenchmark protocol

This appendix defines a lightweight protocol to empirically validate the projections in Section 4.5.3.

Setup. Simulate $K \in \{35, 70, 100\}$ clients on the controlled LAN. Each client sends a fixed tensor matching the update size (0.54M parameters). Compare a plaintext weighted sum with production SA implementation under identical scheduler settings. Throttle uplinks to 10 and 100 Mbps, respectively.

Variants. Evaluate both the no-dropout and dropout-resilient SA rounds. Record whether the stack requires integer payloads and, if so, the packing or fixed-point format applied.

Metrics. Over five runs per condition, the following were recorded: (i) per-client upload bytes, (ii) client masking time, (iii) server unmasking time, and (iv) round duration. The median and interquartile range were reported.

Expectation. For MB-scale updates, communication expands by roughly 1.6-1.8 $\times$ with a modest compute overhead at the client and a small additional server-aggregation cost. Synchronization contributes a small fixed addition per round, which is determined by participation and timeouts.

D. Backdoor results

This appendix summarizes the moderate federated backdoor setting. We report the clean Macro F1, Accuracy, and Attack Success Rate (ASR) in Table D.1.

D.1. Federated backdoor poisoning

The moderate backdoor probe reduced the CICIoMT clean Macro-F1 to 0.9098 because sample-weighted aggregation amplified large malicious CICIoMT clients, and the feature-offset triggered the modality-specific head toward the target pattern, which disproportionately affected the multiclass Macro-F1.

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